

Associated Legendre Polynomial Derivative Methods

Maxwell Idler

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Equation 1.

$$\begin{aligned}\frac{d^m}{dx^m} P_n(x) &= \frac{1}{2^n} \sum_{k=\lceil \frac{n}{2} \rceil}^n \frac{(-1)^{k+n} (2k)!}{(2k-n)!(n-k)!k!} \left(\prod_{i=0}^{m-1} (2k-n-i) \right) x^{(2k-n-m)} \\ &= \frac{1}{2^n n!} \sum_{k=\lceil \frac{n}{2} \rceil}^n (-1)^{k+n} \binom{n}{k} (2k)_{n+m} x^{(2k-n-m)},\end{aligned}$$

where

$$(n)_m = n \cdot (n-1) \cdots (n-m+1).$$

Equation 2.

$$P_n^m(x) = (-1)^m (1-x^2)^{\frac{m}{2}} \frac{d^m}{dx^m} P_n(x)$$